Deep Neural Emulator of Radiative Transfer Model for Near-Infrared Hyperspectral Simulation: Implications for Fast Satellite Retrieval of XCO_2

Ruohua Hu, Tongwen Li, *Member, IEEE*, Jingan Wu, Yuan Wang, and Lingfeng Zhang

***Abstract*—Carbon dioxide (CO_2), a major greenhouse gas, has a profound impact on global climate change. Satellite remote sensing is a crucial approach for monitoring the column-averaged dry-air mole fraction of CO_2 (XCO_2). Currently, XCO_2 retrieval primarily relies on fully physical algorithms, where the radiative transfer model (RTM) imposes a significant computational burden during the forward simulation process. This inefficiency limits the capability for near-real-time monitoring, especially for large-scale retrievals expected from next-generation wide-swath carbon satellites. To address this challenge, this study proposes a deep learning-based surrogate modeling approach to accelerate RTM simulations and enable fast and accurate spectral radiance prediction, thereby facilitating efficient XCO_2 retrieval. Specifically, we design a two-stage neural network (NN) surrogate model for the SCIATRAN RTM, combining an autoencoder (AE) and a deep NN. The former effectively reduces the dimensionality of hyperspectral data while preserving spectral fidelity, and the latter captures complex nonlinear mapping between atmospheric/surface states and satellite-observed radiance under diverse spatiotemporal scenarios. Results demonstrate that the surrogate model significantly improves computational efficiency while maintaining high accuracy, achieving millisecond-level prediction speeds. The model performs robustly across test datasets, finer-step extrapolation scenarios, and real-conditions validation using Orbiting Carbon Observatory-2 (OCO-2) observations, yielding R^2 values close to 1.00 and a minimum root mean square error (RMSE) of 2.87×10^{-4} W/(m^2·sr·nm). Meanwhile, we extend the surrogate model to the calculation of Jacobian matrices of atmospheric and surface parameters, demonstrating high fidelity and stability. Preliminary results demonstrate the potential of the surrogate model for XCO_2 retrieval when embedded within the atmospheric CO2 observations from space (ACOS) retrieval algorithm. This study provides a novel and efficient solution for high-speed XCO_2 retrieval, supporting real-time CO_2 monitoring and carbon flux estimation.**

***Index Terms*—Carbon dioxide, radiative transfer forward simulation, SCIATRAN, surrogate model, XCO_2 retrieval.**

I. Introduction

ATMOSPHERIC carbon dioxide (CO_2) concentrations have been continuously increasing since the Industrial Revolution. As of December 2023, the global average column-averaged dry-air mole fraction of CO_2 (XCO_2) had reached 420.0 ppm, which is approximately 151% of preindustrial levels [1]. The persistent rise in CO_2 levels has contributed to a wide range of adverse environmental impacts [2], including global warming, sea level rise, ocean acidification, glacier retreat, and an increased frequency of extreme weather events such as floods, droughts, tropical cyclones, and wildfires [3], [4]. Therefore, accurate and timely monitoring of atmospheric CO_2 is of critical importance. Compared to ground-based observations, satellite remote sensing offers global coverage [5], [6], making it a powerful tool for monitoring XCO_2 on a global scale [7]. In recent years, several satellites dedicated to atmospheric CO_2 measurements have been launched, including Orbiting Carbon Observatory-2 (OCO-2), OCO-3, GOSAT, GOSAT-2, and TanSat [8], [9]. These satellites have generated global XCO_2 datasets that are extensively used for monitoring XCO_2 spatiotemporal variability and estimating carbon sources and sinks [10].

Currently, satellite-based retrievals of XCO_2 primarily rely on fully physical algorithms [11], [12], [13], which usually consist of two core components: radiative transfer forward simulation and iterative optimization for inversion [14], [15]. The forward process refers to the use of a radiative transfer model (RTM) to simulate the top-of-atmosphere radiance received by the satellite under specific viewing geometries based on inputs such as atmospheric state variables and surface albedo. The inverse process iteratively adjusts the input state parameters by minimizing the discrepancy between the simulated spectra and the satellite observations, thereby accurately retrieving atmospheric XCO_2. Fig. 1 illustrates the basic workflow of XCO_2 retrieval. Solar radiation passes through the

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Ruohua Hu is with the School of Geospatial Engineering and Science, Sun Yat-sen University, Zhuhai 519082, China (e-mail: hurh6@mail2.sysu.edu.cn).

Tongwen Li and Jingan Wu are with the School of Geospatial Engineering and Science and the Key Laboratory of Comprehensive Observation of Polar Environment, Ministry of Education, Sun Yat-sen University, Zhuhai 519082, China (e-mail: litw8@mail.sysu.edu.cn; wujg5@mail.sysu.edu.cn).

Yuan Wang is with the School of Atmospheric Physics, Nanjing University of Information Science and Technology, Nanjing 210044, China (e-mail: 003785@nuist.edu.cn).

Lingfeng Zhang is with the Department of Atmospheric and Oceanic Sciences, Fudan University, Shanghai 200433, China, and also with the Key Laboratory of Radiometric Calibration and Validation for Environmental Satellites, National Satellite Meteorological Center (National Center for Space Weather), Innovation Center for FengYun Meteorological Satellite, Beijing 100081, China (e-mail: lfzhang24@m.fudan.edu.cn).

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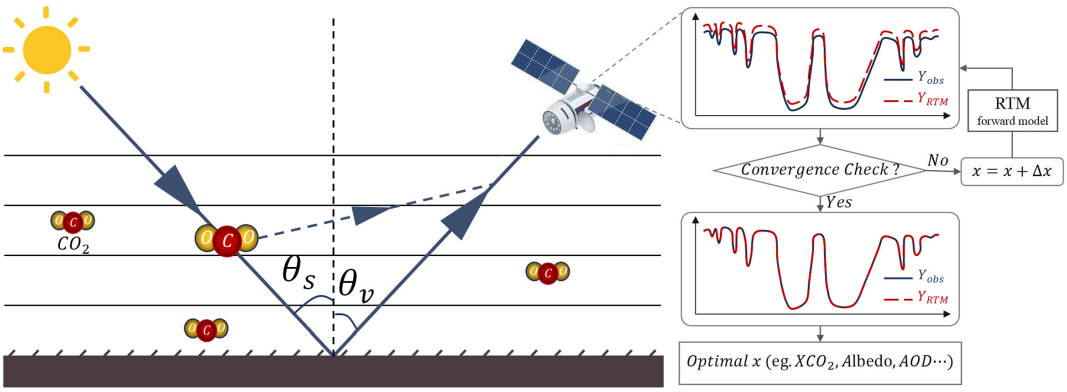

Fig. 1. Basic workflow of XCO_2 retrieval. θ_s and θ_v represent the SZA and the viewing zenith angle, respectively. Y_{obs} and Y_{RTM} denote the observed spectral radiance and the RTM-simulated spectral radiance, respectively. AOD means the aerosol optical depth.

atmosphere, reflects off the surface, and is partially absorbed and scattered by CO_2, aerosols, and other constituents before reaching the satellite. The observed spectral radiance is then compared with that simulated by RTM, and the residuals guide iterative parameter updates [16], [17]. However, this approach often relies on high-resolution hyperspectral forward simulations in the near-infrared [18], [19], resulting in low computational efficiency. Combined with the iterative inversion, this leads to considerable computational overhead and extremely slow processing speeds [20], [21], [22]. These limitations hinder the large-scale near-real-time CO_2 monitoring and CO_2 flux estimation, a challenge further exacerbated by next-generation carbon satellites with significantly wider swath widths [23], [24], [25].

Artificial intelligence (AI) has demonstrated considerable potential in enhancing computational efficiency through its capability for rapid prediction. A current line of research directly leverages satellite spectral measurements and XCO_2 products or ground-truth data to train machine learning models. For instance, David et al. [26] employed an artificial neural network (ANN) to retrieve XCO_2 from OCO-2 spectra, using XCO_2 products from the Copernicus Atmosphere Monitoring Service (CAMS) as the training target. Their results demonstrated retrieval accuracy comparable to that of conventional physics-based algorithms, with an approximately 20% reduction in systematic bias. Bréon et al. [27] further refined the input feature set, thereby improving the model's sensitivity to localized emission enhancement signals. Nevertheless, given the continuously increasing trend in atmospheric XCO_2, AI models trained solely on historical data often struggle to achieve accurate future retrievals. To address this limitation, some studies have incorporated simulated data generated by RTM to augment training datasets and simulate higher-concentration scenarios. For example, Xie et al. [28] constructed a multilayer perceptron model trained on both real OCO-2 observations and synthetic spectra generated by the ReFRACtor RTM, with both OCO-2 L2 XCO_2 retrievals and simulated XCO_2 inputs as targets. The resulting model exhibited strong spatiotemporal consistency in XCO_2 predictions over East Asia. While such AI-based approaches offer substantial improvements in computational efficiency, they often model the relationship between XCO_2 and satellite observations solely to retrieve XCO_2 data, and the absence of uncertainty information, such as averaging kernels, may limit their applicability in carbon assimilation systems.

Another particularly promising direction involves the construction of AI-based surrogate models that emulate radiative transfer forward simulation. By integrating these surrogate models into existing physically based retrieval frameworks, it becomes possible to substantially accelerate the retrieval process without compromising retrieval accuracy and physical interpretability. Unlike purely data-driven methods, these approaches are grounded in physical modeling, thereby preserving theoretical rigor, interpretability, and uncertainty information. This approach has been previously explored in quantitative remote sensing, notably in land surface temperature retrieval [29], [30]. Specifically, the RTM is used to impose physical constraints, while AI serves as a surrogate model for computational optimization [29]. For example, Liu et al. [31] proposed a PCA-based spectral compression method, while Le et al. [32] employed a neural network (NN) for fast spectral emulation and demonstrated superior performance compared with PCA. These studies further highlight the potential of AI-based surrogate models in improving the efficiency of radiative transfer computations. This hybrid architecture preserves the physical interpretability while significantly improving computational efficiency.

Despite its promise, the application of surrogate modeling to XCO_2 retrieval still faces several critical challenges. On the one hand, XCO_2 retrieval commonly relies on near-infrared hyperspectral observed spectra, where the input contains several variables, while the output corresponds to spectra with thousands of channels. This results in a highly complex nonlinear mapping, placing considerable demands on the learning capacity and generalization ability of AI models. On the other hand, satellite observations are inherently conducted under highly heterogeneous spatiotemporal scenarios, where variations in viewing geometry, atmospheric state, and surface properties can significantly influence the radiative transfer process. Effectively addressing the challenges associated with hyperspectral simulation in complex and dynamic scenarios,

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TABLE I
INPUT PARAMETER RANGES USED IN THE RADIATIVE TRANSFER FORWARD SIMULATION

Parameter		Predefined Range	Real Conditions (Primary Range)	Precision /Resolution
Geographical Information	Latitude	[-90°,90°]	[-80°,80°]	0.1°
	Longitude	[0°,360°]	[0°,360°]	0.1°
	SZA	[0°, 90°]	[22°, 50°]	0.1°
	RAA	[0°, 180°]	[170°, 180°]	0.1°
	VA	[0°, 75°]	[15°, 40°]	0.1°
Atmosphere / Surface Information	AOD	[0, 1]	[0, 0.2]	0.001
	XCO_2	[380ppm, 430ppm]	[404ppm, 415ppm]	0.1ppm
	Albedo	[0, 1]	[0.003, 0.2]	0.001
	Surface pressure	[980hPa, 1100hPa]	[980hPa, 1100hPa]	1hPa
Profile Information	Pressure profile Temperature profile Altitude profile	B2D files corresponding to specific months and latitudes		

and leveraging AI-based surrogate models to emulate the radiative transfer process, constitutes a critical step toward high-efficiency XCO_2 retrieval.

Therefore, this study adopts SCIATRAN, a high-spectral-resolution RTM developed by Rozanov et al. [33] at the University of Bremen, as the physical reference, and a two-stage NN architecture is proposed to construct a surrogate model that emulates SCIATRAN's spectral radiance outputs, with the goal of significantly accelerating XCO_2 retrieval from satellite observations. The proposed model first employs an autoencoder (AE) to effectively reduce the dimensionality of the high-resolution spectra. In the second stage, a hybrid NN architecture is constructed by integrating a deep fully connected NN (DNN) with a long short-term memory (LSTM) network, which is designed to capture the complex nonlinear mapping between atmospheric state variables and surface state variables and the satellite-observed radiance under diverse spatiotemporal scenarios. Furthermore, this study extends the surrogate model for Jacobian matrix generation and XCO_2 retrieval. Specifically, the spectra-Jacobian surrogate model is embedded into a full-physics inversion framework, achieving comparable accuracy while substantially accelerating simulation speed and reducing computational cost. The proposed approach holds promise for providing efficient and near-real-time support for satellite XCO_2 retrieval applications.

II. DATA AND METHODS

A. Data

1) OCO-2 Products: The OCO-2 is an Earth observation satellite mission operated by the National Aeronautics and Space Administration (NASA), which was successfully launched on July 2, 2014. The onboard instrument employs a push-broom scanning technique, with a ~10-km swath and spatial resolution of 1.29 km (cross-track) ×2.25 km (along-track), enabling precise greenhouse gas measurements. It observes three spectral bands: one for the oxygen absorption window and two for carbon dioxide. Specifically, the O2 A-band spans 0.76 μm (near-infrared, NIR), while CO_2 absorption is measured at 1.6 (WCO_2) and 2.0 μm(SCO_2). Each spectral band contains 1016 high-resolution spectral channels, allowing for detailed characterization of solar radiance after atmospheric transmission.

In this study, we primarily utilize the OCO2_L1B_Science v11r data product [34], which provides calibrated spectral radiance in the WCO_2 band, along with instrument line shape (ILS) parameters (e.g., ils_delta_lambda and ils_relative_response) and dispersion coefficients (dispersion_coef_samp) for further response function correction and spectral calibration. In addition, the OCO2_L2_Standard v11 data product from the same observation period is used to extract key retrieved variables, including XCO_2, aerosol optical depth (AOD) at 755 nm, surface albedo in the WCO_2, observation geometry parameters, and geolocation information. It should be noted that our study focuses on ocean regions, where surface albedo can be reasonably assumed to be spectrally constant within the WCO_2 band (1588–1624 nm). For future land-surface applications, where albedo varies significantly with wavelength, the model architecture should be extended to include wavelength-dependent albedo vectors as inputs.

2) Radiative Transfer Simulation Data: SCIATRAN is a high-spectral-resolution RTM developed by the University of Bremen. It solves the radiative transfer equation to simulate the propagation of both solar and terrestrial radiation through the atmosphere, enabling highly accurate forward simulations. One of SCIATRAN's key strengths lies in its ability to model the influence of various atmospheric constituents, such as trace gases and aerosols, on electromagnetic radiation, including scattering, absorption, and emission processes. Moreover, SCIATRAN incorporates Earth's curvature and atmospheric refraction effects, allowing for a more realistic representation of radiative energy transmission paths and spatial distributions in the atmosphere. Thus, this study employs SCIATRAN to construct a synthetic dataset for surrogate model training. In addition, the WCO_2 spectral band is selected for XCO_2 retrieval due to its ability to deliver high retrieval accuracy while reducing dependence on complex atmospheric scenarios, thereby simplifying the inversion process and improving consistency with ground-based measurements [35], [36].

Table I summarizes the input parameters and their corresponding value ranges. The predefined range represents

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the theoretical or physically allowable limits of each input parameter used in the forward radiative transfer simulations, while the primary range denotes the typical range of parameter values observed under real atmospheric and surface conditions. Although input parameters can theoretically be selected across the predefined range, many combinations within these idealized ranges rarely occur under real atmospheric or surface conditions, and relying solely on such ideal values may reduce the realism of the simulations [28]. In this study, we generate samples from both ranges in approximately a 1:1 ratio. In this way, a greater emphasis is placed on the primary range to better capture real-world variability during model training because more than half of the samples are concentrated within the primary range, which represents a relatively narrower but more realistic subset of the entire parameter space. The primary range is derived from OCO-2 observations in 2020. To enhance the generalization ability of the surrogate model and ensure robust performance under highly heterogeneous spatiotemporal scenarios, multiple combinations of input parameters are randomly sampled within these ranges. The sampled parameters are then used to drive SCIATRAN's forward simulations in the WCO_2 spectral band (1588–1624 nm), with a spectral sampling interval of 0.025 nm (see Section III-C1), resulting in 1441 spectral channels per simulation. It is important to note that, given the complexity of the XCO_2 retrieval algorithm, this study focuses on ocean regions where surface conditions are relatively uniform for both the simulation and retrieval experiments. In principle, this approach can be extended to land regions in future applications. In total, 13 766 synthetic spectral samples are generated for model training and analysis. The top-of-atmosphere radiance at wavelength λ (L_λ^{TOA}), simulated by SCIATRAN, can be formally expressed as

$$L_\lambda^{\mathrm{TOA}} = \mathrm{RTM}\,(\lambda, \mathrm{AOD}, \mathrm{XCO_2}, \mathrm{Albedo}, \mathrm{Pressure}\ \mathrm{Latitude}, \mathrm{Longitude}, \mathrm{SZA}, \mathrm{RAA}, \mathrm{VA}\ldots) \tag{1}$$

where RTM(·) denotes the forward simulation by the SCIATRAN, which accounts for key physical variables including AOD, surface albedo, XCO_2, surface pressure, and geometric observation angles such as solar zenith angle (SZA), viewing angle (VA), and relative azimuth angle (RAA).

3) Auxiliary Data: In addition to the OCO-2 data, this study incorporates the TSIS-1 hybrid solar reference spectrum (HSRS), provided by LaTiS, as the reference input for solar irradiance [37]. With a spectral resolution of up to 0.025 nm, HSRS offers highly accurate solar spectral data essential for radiative transfer forward simulations. For atmospheric background settings, we adopt the default atmospheric profiles from the Bremen 2-D (B2D) chemical transport model, developed by the University of Bremen. Based on this model [38], vertical distributions of trace gases (e.g., CO_2) are specified and used as atmospheric composition inputs in the radiative transfer forward simulations.

B. Methodology

This study aims to develop a deep learning-based surrogate model of SCIATRAN to improve the computational efficiency of satellite-based XCO_2 retrieval. To address the challenge of simulating high-dimensional spectra, a high-fidelity dimensionality reduction module is designed within the deep learning framework. Furthermore, to cope with the complexity of diverse spatiotemporal scenarios, a synthetic dataset with broad spatiotemporal coverage is constructed, and a deep NN module is developed to capture the nonlinear mapping between the input atmospheric and surface states and the output spectral radiance. The overall methodology consists of three main stages, as illustrated in Fig. 2.

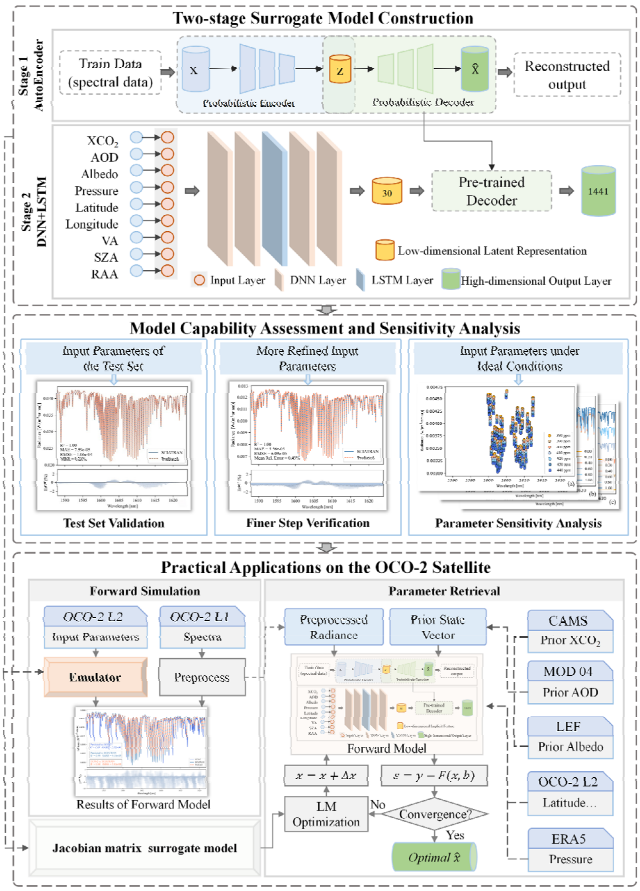

Fig. 2. Overview of the proposed framework.

1) Surrogate Model Construction: A two-stage deep learning architecture is developed, combining AE and DNN-LSTM. The AE is used to extract and compress spectral features, while the DNN-LSTM module captures the nonlinear mapping between input parameters and output spectral radiance.

2) Model Capability Assessment and Sensitivity Analysis: To verify the surrogate model's extrapolation capability, a finer-step verification is conducted using test sets generated with more refined input parameters than those used during training (e.g., refining the XCO_2 interval from 0.1 to 0.01 ppm). In addition, the model's sensitivity to key input parameters is quantitatively analyzed.

3) Application to Real Satellite Data: The forward simulation accuracy of the surrogate model is tested under OCO-2 observational scenarios. Furthermore, the proposed framework is extended to simulate Jacobian matrices (see Section IV-A),

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TABLE II
NETWORK ARCHITECTURE AND HYPERPARAMETERS OF THE SURROGATE MODEL

Model	Component	Architecture/Configuration
AutoEncoder (AE)	Encoder	4 fully connected layers
	Decoder	4 fully connected layers
	Activation function	ReLU
	Latent dimension	30
	Loss function	MSE
DNN-LSTM Forward Model	DNN	3 fully connected layers
	LSTM	1 layer (hidden size = 256)
	Output layers	3 fully connected layers
	Loss function	MSE

and then, both the spectral and Jacobian surrogate models are integrated into the Atmospheric CO_2 Observations From Space (ACOS) retrieval framework [17], enabling fast XCO_2 inversion. Details are as follows:

1) Two-Stage Surrogate Model Construction: To efficiently map complex atmospheric and surface parameters to radiative outputs in the WCO_2 absorption bands, this study develops a surrogate modeling framework that integrates radiative transfer forward simulations with deep learning. As illustrated in part 1 of Fig. 2, the framework consists of two main components: dimensionality reduction and nonlinear feature mapping. The mapping from physical parameters to high-dimensional radiance spectra is expressed as

$$\mathrm{y} = f\left(g\left(x\right)\right) \tag{2}$$

where x denotes the input physical parameters, $g(\cdot)$ represents a DNN-LSTM with the output dimension of 30, and $f(\cdot)$ denotes the decoder within the AE that reconstructs the full spectral output, represented by y. It consists of 1441 spectral channels covering the WCO_2 absorption band. The detailed layer configuration, activation functions, and loss functions for both the AE and DNN–LSTM modules are summarized in Table II.

Since the spectral output consists of a high-dimensional continuous spectrum, directly modeling the full spectrum may result in issues such as the curse of dimensionality and redundancy interference. To address this, all spectral data are first normalized to eliminate scale disparities and enhance training stability. Then, an AE is employed to perform nonlinear dimensionality reduction on the spectral data. The AE consists of an encoder and a decoder: the encoder extracts a 30-D latent representation that preserves the principal absorption features while filtering out redundant components; the decoder subsequently reconstructs the full radiance spectra from this low-dimensional latent space.

For nonlinear feature mapping, nine physical variables are selected as model inputs, including AOD, XCO_2, surface albedo, surface pressure, geographic coordinates (latitude and longitude), and geometric parameters (VA, SZA, and RAA). Prior to training, both input parameters and spectral data are standardized using the StandardScaler implemented in scikit-learn, which normalizes each variable to have zero mean and unit variance, ensuring consistency with the preprocessing applied to the spectral data. A hybrid network architecture is then constructed by combining DNN with LSTM. The DNN performs a nonlinear transformation and abstraction of the normalized input features. Subsequently, the LSTM module is employed to capture the sequential dependencies across spectral bands, enhancing the model's ability to represent complex spectral structures. The output of the DNN-LSTM structure is a 30-D latent representation that encodes the nonlinear mapping from atmospheric and surface inputs to spectral radiance characteristics. The decoder subsequently reconstructs the full 1441-D spectral output from this representation. This two-stage architecture, comprising spectral dimensionality reduction and nonlinear feature mapping, effectively integrates the advantages of feature encoding and sequence modeling. It ensures high-fidelity reconstruction of spectral radiance while bridging the dimensional gap between input parameters and spectral outputs, thereby providing an efficient solution for radiative transfer simulation in the WCO_2 absorption band.

2) Model Capability Assessment and Sensitivity Analysis: The accuracy and stability of the surrogate model are quantitatively assessed by comparing its radiance output with that of the SCIATRAN model. The evaluation consists of two main approaches: 1) assessing performance on a hold-out subset of the synthetic dataset and 2) evaluating the model's performance in simulating scenarios with more refined input parameters. We adopt several statistical metrics to characterize model performance, including mean absolute error (MAE), root mean square error (RMSE), coefficient of determination (R^2), and mean relative error (MRE), which are defined as follows:

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|x_{\mathrm{true},i} - x_{\mathrm{pred},i}\right| \tag{3}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(x_{\mathrm{true},i} - x_{\mathrm{pred},i}\right)^2} \tag{4}$$

$$R^2 = \left[\frac{\sum_{i=1}^{N}\left(x_{\mathrm{true},i}-\overline{x_{\mathrm{true}}}\right)\left(x_{\mathrm{pred},i}-\overline{x_{\mathrm{pred}}}\right)}{\sqrt{\sum_{i=1}^{N}\left(x_{\mathrm{true},i}-\overline{x_{\mathrm{true}}}\right)^2}\sqrt{\sum_{i=1}^{N}\left(x_{\mathrm{pred},i}-\overline{x_{\mathrm{pred}}}\right)^2}}\right]^2 \tag{5}$$

$$\mathrm{MRE} = \frac{1}{N}\sum_{i=1}^{N}\frac{\left|x_{\mathrm{true},i} - x_{\mathrm{pred},i}\right|}{x_{\mathrm{true},i}} \tag{6}$$

where x_{true} and x_{pred} denote the spectral radiance value of the truth and the prediction, respectively; $\overline{x}$ represents the corresponding mean values, and N is the number of spectral channels. Notably, when calculating the overall performance across multiple spectral samples, each radiance spectrum is first flattened into a 1-D array before statistical analysis is performed. Based on the trained surrogate model, we further conduct a sensitivity analysis to investigate how key variables, such as XCO_2, surface albedo, and AOD, affect satellite-observed radiance at the 1.6-μm band.

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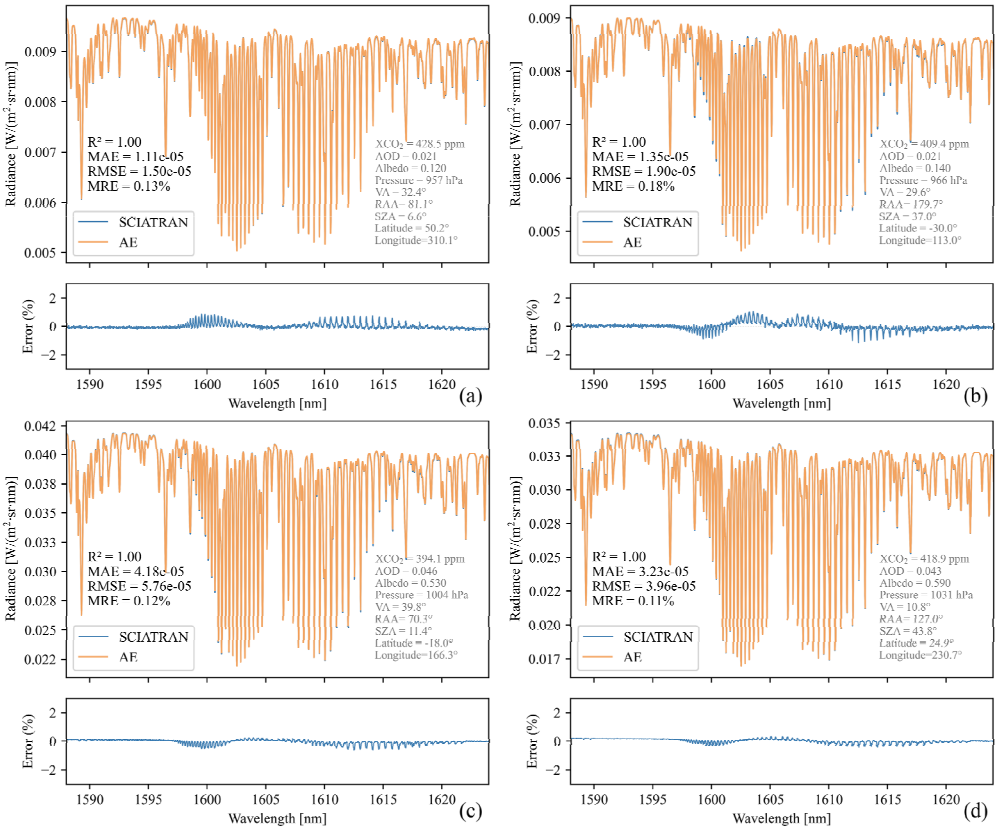

Fig. 3. Evaluation of radiance spectrum reconstruction using the AE model. (a)–(d) Four distinct simulation scenarios.

3) Practical Applications on the OCO-2 Satellite: The real-world application of the surrogate model to the OCO-2 satellite involves both forward simulation and XCO_2 retrieval. Specifically, the surrogate-generated spectra are first compared with actual OCO-2 observations to assess the model's capability in reproducing radiance spectra under real scenarios. Subsequently, to preliminarily assess its feasibility for satellite retrieval of XCO_2, we embed the surrogate forward model into the ACOS retrieval framework, which is a comprehensive physical retrieval system originally developed by NASA's OCO project [17], [39], and it has been successfully applied to XCO_2 retrievals from the OCO-2/3 and GOSAT missions [40], [41].

The retrieval process, based on the integration of the surrogate model into the ACOS framework, consists of two main steps. First, the surrogate model is used to generate simulated spectra based on given input parameters $(\vec{x}, \vec{b})$. Second, an optimization algorithm is applied to retrieve the most probable target state vector $(\vec{x})$ by minimizing the cost function [15]

$$\chi^2 = \left(\vec{y} - \vec{F}\left(\vec{x}, \vec{b}\right)\right)^T S_\varepsilon^{-1}\left(\vec{y} - \vec{F}\left(\vec{x}, \vec{b}\right)\right) + \left(\vec{x} - \vec{x_0}\right)^T S_\alpha^{-1}\left(\vec{x} - \vec{x_0}\right) \quad (7)$$

where $\vec{y}$ is the observed radiance, $\vec{F}(\vec{x}, \vec{b})$ is the forward-simulated spectra produced by the surrogate model, $\vec{x_0}$ is the a priori state vector, and S_ε and S_α are the covariance matrices for measurement error and prior uncertainty, respectively. The auxiliary vector $\vec{b}$ includes fixed or known parameters such as observation geometry and geographic location. To solve this nonlinear retrieval problem, the Levenberg–Marquardt (LM) method is employed for iterative optimization

$$\vec{x}_{i+1} = \vec{x}_i + \widehat{S}_i\left[K_i^T S_\varepsilon^{-1}\left(\vec{y} - \vec{F}\left(\vec{x}_i, \vec{b}\right)\right) - S_\alpha^{-1}\left(\vec{x}_i - \vec{x_0}\right)\right] \quad (8)$$

$$\widehat{S}_i = \left[K_i^T S_\varepsilon^{-1} K_i + (1+\gamma) S_\alpha^{-1}\right]^{-1} \quad (9)$$

where $K_i = (\partial\vec{F})/(\partial\vec{x}_i)$ is the Jacobian matrix, presenting the sensitivity of the radiance spectrum to changes in state variables. To accelerate this step, we additionally develop a surrogate model for the Jacobian matrix using the same input variables, enabling rapid retrievals while maintaining retrieval accuracy (see Section IV-A).

III. Results

A. Accuracy Evaluation of the Surrogate Model

1) AE Performance: To assess the standalone performance of the AE, the dataset is subjected solely to dimensionality reduction and reconstruction without surrogate modeling. A total of 13 766 samples is split into training and test sets at a 4:1 ratio. On the test set, the AE demonstrates excellent reconstruction performance, with the reconstructed spectra closely matching the original spectra. In terms of quantitative metrics, MAE, RMSE, and R^2 reach values of 4.64×10^{-5} W/(m^2·sr·nm), 8.86×10^{-5} W/(m^2·sr·nm), and 1.00, respectively, indicating consistently minimal reconstruction error. Fig. 3 illustrates representative reconstruction results of the

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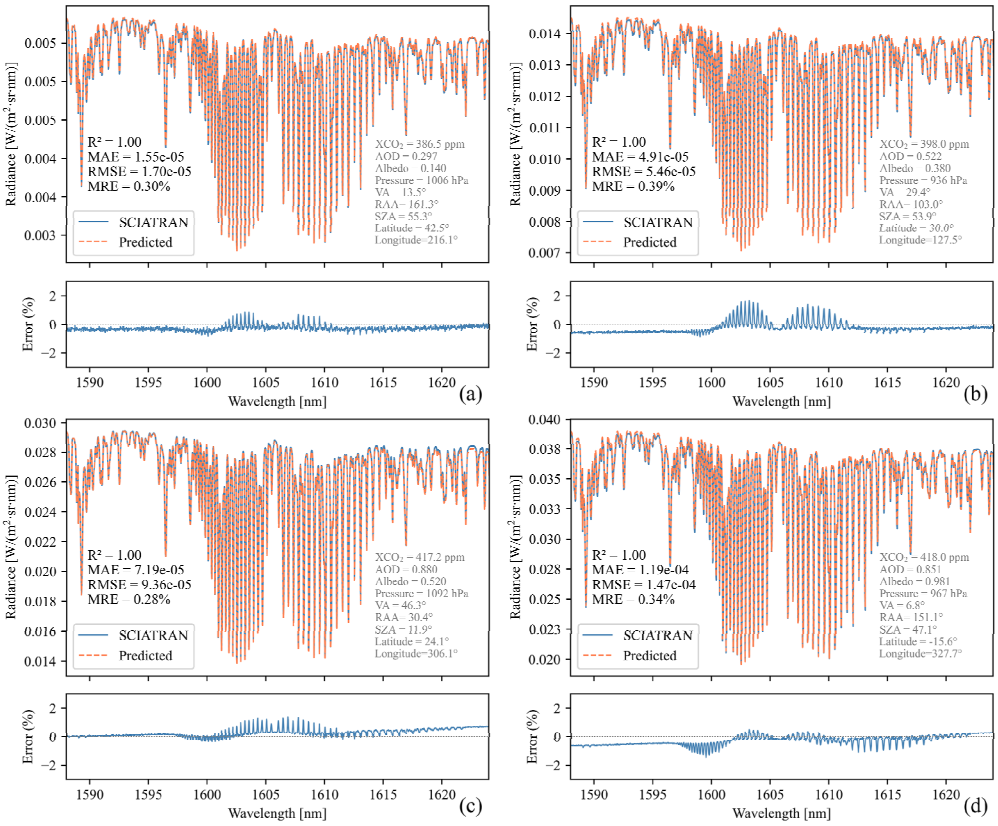

Fig. 4. Evaluation of radiance spectrum reconstruction using the surrogate model. (a)–(d) Four distinct simulation scenarios.

AE for selected test samples. Across different input samples, the RMSE and MAE remain in the order of 10^{-5}, with the MRE approaching 0 and R^2 values close to or exactly equal to 1, confirming the AE's high spectral fidelity. These results demonstrate that the AE model achieves effective spectral compression with high precision, providing a reliable dimensionality reduction strategy and a robust latent representation for subsequent surrogate model development.

2) Surrogate Model Accuracy: Consistent with the previous setup, the dataset is split into training and test sets at a 4:1 ratio. The surrogate model also demonstrates high overall accuracy on the test set, with an MAE of 2.37×10^{-4} W/(m^2·sr·nm), an RMSE of 4.21×10^{-4} W/(m^2·sr·nm), and an R^2 of 0.9994. To further evaluate the model's performance in reconstructing radiance spectra, several representative samples are selected for both qualitative and quantitative analyses. The predicted spectra are compared with the reference spectra from SCIATRAN, with corresponding accuracy metrics reported in Fig. 4. Results show that the surrogate model effectively captures the spectral shape across most cases, with predicted curves closely aligning with SCIATRAN outputs. In particular, all selected samples yield R^2 values close to 1.00, indicating excellent spectral agreement. Both MAE and RMSE remain at low levels, and MREs are consistently below 2%, confirming the model's robustness and generalization ability under varying input conditions. These results demonstrate that the proposed surrogate modeling framework is capable of reconstructing spectral radiance with high precision while preserving fine spectral structures.

B. Application of the Surrogate Model to Finer-Step Simulations

To further evaluate the extrapolation capability of the surrogate model, this study refines the input parameters by setting smaller step sizes and conducts a comparative analysis between the SCIATRAN simulations and the surrogate model predictions under these finer-step inputs. Specifically, while the training data employ a sampling interval of 0.1 for variables such as SZA, VA, RAA, and XCO_2, the extrapolation dataset refines these intervals to 0.01. Similarly, the sampling intervals for AOD and albedo are refined from 0.01 in the training set to 0.001 in the extrapolation set.

Using an extrapolation dataset of 1200 finer-step samples, the overall simulation results of the surrogate model show excellent performance, with an MAE of 3.89×10^{-4} W/(m^2·sr·nm), an RMSE of 4.92×10^{-4} W/(m^2·sr·nm), and an R^2 of 0.9995. Representative reconstruction results are shown in Fig. 5, demonstrating that the surrogate model maintains high predictive accuracy even under input conditions with finer sampling intervals. These results further confirm the model's ability to accurately simulate spectra under more demanding and fine-scale scenarios. Such a strategy, refining sampling intervals of input parameters to improve spectral simulation accuracy, proves

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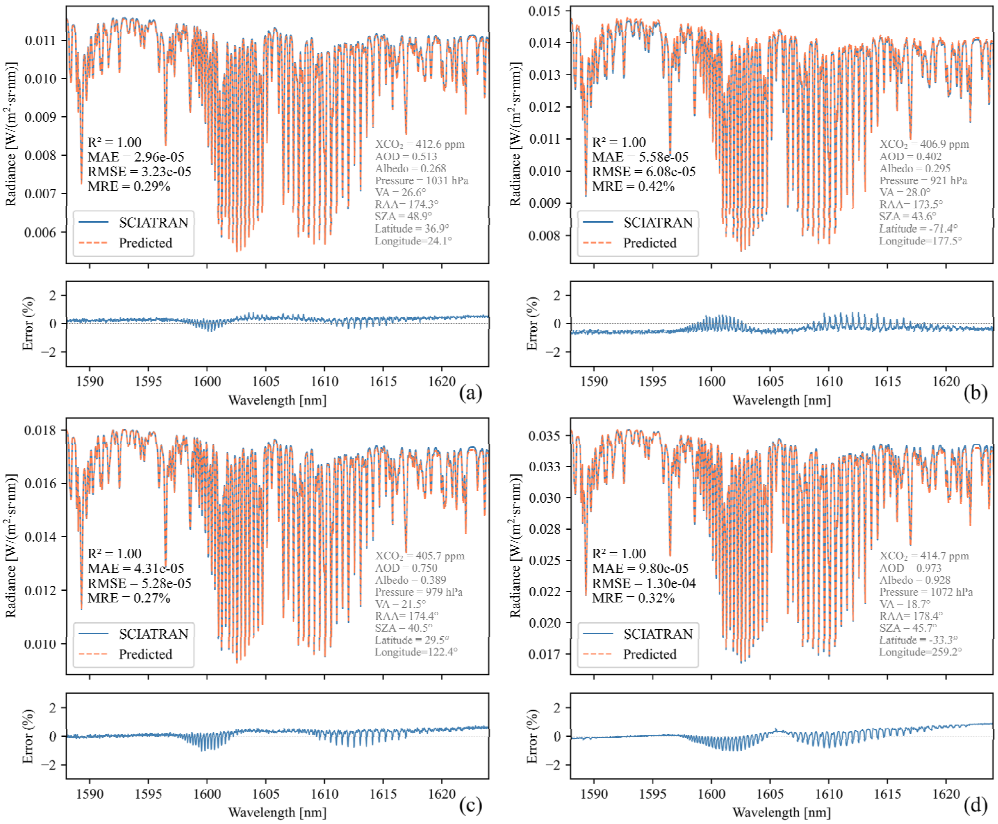

Fig. 5. Surrogate model performance under finer-step input conditions. (a)–(d) Four distinct simulation scenarios.

to be an effective approach for supporting the generation of fine-grained spectral lookup tables.

C. OCO-2 Satellite Application

1) Data Preprocessing: Spectral dispersion in OCO-2 observations can vary slightly due to several factors, including Doppler shifts caused by relative motion between the instrument and the reflecting surface, slight displacements between the focal plane array (FPA) [42] and the optical bench assembly (OBA) [42], and grating tilt induced by thermal gradients within the OBA. In addition, manufacturing inconsistencies, instrument aging, and dynamic environmental conditions may lead to deviations between actual instrument responses and their nominal design. Therefore, before evaluating the performance of SCIATRAN simulations for WCO_2 under OCO-2 observational conditions, spectral calibration is conducted using the OCO-2 wavelength dispersion equation [see (10)], followed by radiometric correction via the instrument response function [see (11)]

$$\lambda = \sum_{i=0}^{5} c_i \cdot \text{column}^i \tag{10}$$

$$F_{\text{calibrated}} = \frac{\int_0^\infty F_0(\lambda)\,\Gamma(\lambda)\,d\lambda}{\int_0^\infty \Gamma(\lambda)\,d\lambda} \tag{11}$$

where λ denotes the calibrated wavelength, c represents the dispersion coefficients, and column refers to the spectral channel index (ranging from 1 to 1016). F_0 and $F_{\text{calibrated}}$ represent the original and calibrated spectral radiance, respectively, and $\Gamma(\lambda)$ denotes the instrument response function.

The spectral sampling interval used in SCIATRAN forward simulations directly affects the precision and fidelity of the output spectra. Thus, taking the OCO-2 sounding with Sounding_ID 2020010118141401 as a case study, we investigate the discrepancies between SCIATRAN simulations with varying sampling intervals and the corresponding OCO-2 observations (see Fig. 6). When the sampling interval is finer than 0.025 nm, the simulation accuracy remains stable, with the RMSE and MRE maintained at 1.01×10^{-4} W/(m^2·sr·nm) and 2.75%, respectively. In addition, the errors across channels within the WCO_2 band are generally consistent. However, when the interval is coarsened to 0.1 nm, the simulation accuracy deteriorates significantly, with the RMSE rising to 1.53×10^{-4} W/(m^2·sr·nm) and the error in some channels reaching up to −25%, indicating a pronounced underestimation. In other words, a finer spectral sampling interval yields higher spectral resolution and more detailed spectral features though at the cost of increased computational demand and longer runtime. To balance simulation accuracy and efficiency, a spectral sampling interval of 0.025 nm is adopted in this study for the generation of SCIATRAN simulation datasets.

2) Evaluation of Surrogate Model Accuracy on OCO-2 Observations: To assess the surrogate model's adaptability to real satellite observations, this study selects high-quality OCO-2 soundings under clear-sky conditions in January 2020.

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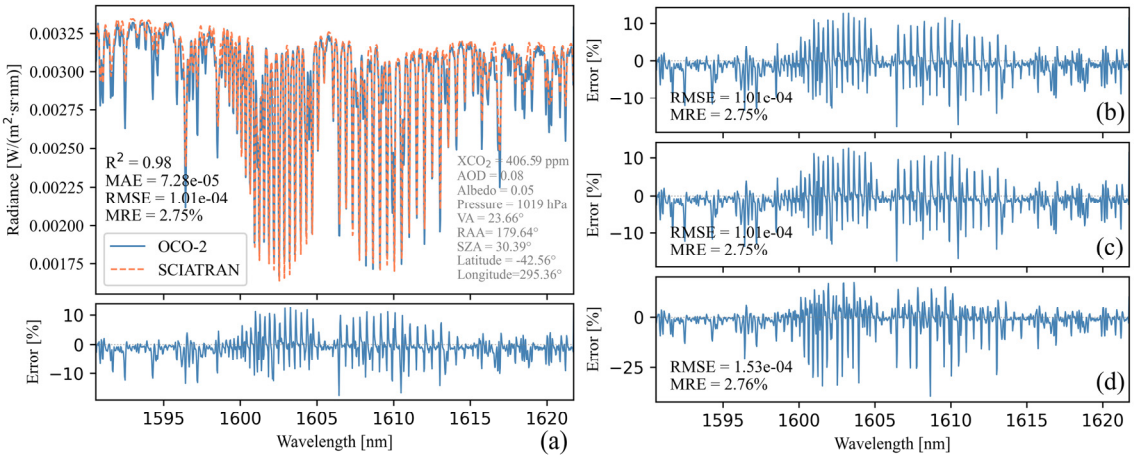

Fig. 6. Comparison between SCIATRAN-simulated spectra and OCO-2 observations under different spectral sampling intervals. (a) 0.025 nm. (b)–(d) Represent resolutions of 0.01, 0.02, and 0.1 nm, respectively.

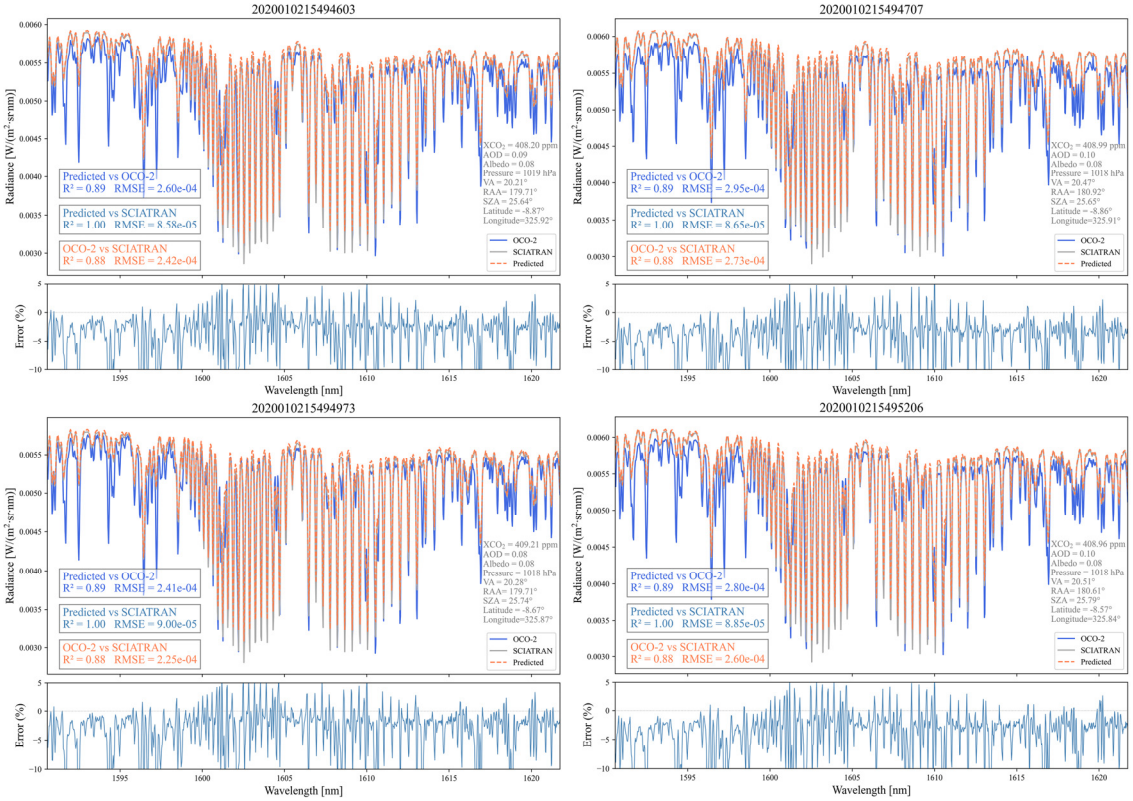

Fig. 7. Surrogate model predictions compared with actual OCO-2 observations.

Based on associated state parameters, the surrogate model is used to predict the corresponding radiance spectra. These predictions are then compared with actual OCO-2 observations. The overall performance of the surrogate model on these real observations yielded an MAE of 2.24×10^{-4} W/(m^2·sr·nm), an RMSE of 2.87×10^{-4} W/(m^2·sr·nm), and R^2 of 0.9769. As shown in Fig. 7, four representative sample points demonstrate that the predicted spectra closely match the observed spectra in both shape and magnitude, accurately capturing the depth and position of major absorption features with the WCO_2 band. The surrogate-model predictions yield MRE of only 0.87%, 0.82%, 1.01%, and 0.86% relative to the SCIATRAN simulations, while the corresponding deviations from the actual OCO-2 observations are 4.01%, 4.69%, 3.63%, and 4.31%, respectively. It is worth noting that, compared to Fig. 4, the results shown in Fig. 7 appear to be less satisfactory. This is primarily because the surrogate model is trained on SCIATRAN-simulated data, resulting in particularly good agreement with the simulation. However, certain discrepancies exist between SCIATRAN simulations and actual satellite observations, which may explain the larger differences observed in Fig. 7. These results confirm the surrogate model's

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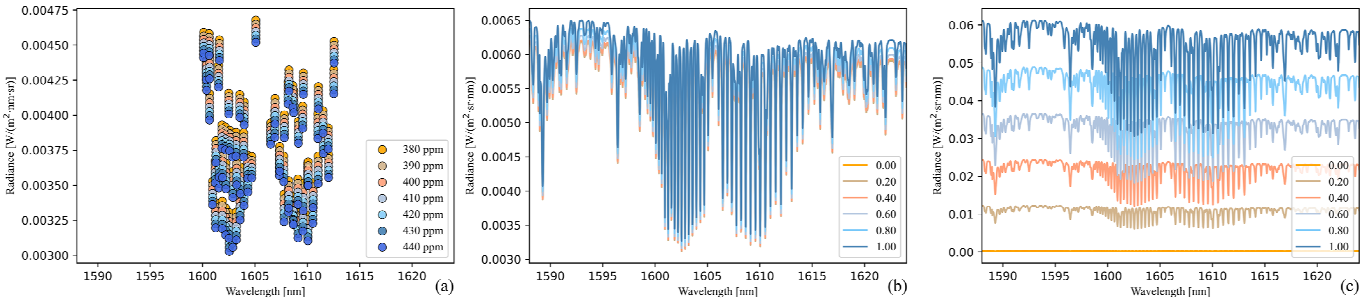

Fig. 8. Sensitivity analysis of the surrogate model for radiance spectra. (a) XCO_2, (b) AOD, and (c) albedo. Default values for XCO_2, AOD, and albedo are set to 410 ppm, 0.1, and 0.1, respectively.

robustness and strong generalization capability when applied to actual OCO-2 observational conditions.

D. Sensitivity Analysis

Observation geometry, including SZA, RAA, and VA, affects the observed radiance by altering the radiation path through the atmosphere. In addition, physical variables such as trace gas concentrations (primarily CO_2 within the WCO_2 band), AOD, and surface albedo influence the processes of atmospheric absorption and scattering, thereby impacting the observed radiance. Fig. 8 illustrates the sensitivity of the surrogate model with respect to these three parameters. As shown, the radiance response exhibits substantial variability depending on the parameter in question. For XCO_2, changes within the current spectral window result in relatively minor variations in radiance, with a high degree of overlap among the corresponding curves. Therefore, Fig. 8(a) highlights only those spectral channels with relatively higher sensitivity using scatter plots. The results present a challenge for retrieval applications, as even substantial perturbations in XCO_2 may not induce significant responses in the observed radiance. Consequently, achieving accurate XCO_2 retrievals may require high signal-to-noise ratio (SNR) observations or joint retrieval with other more sensitive parameters.

In contrast, both AOD and surface albedo exhibit more pronounced spectral responses. An increase in AOD leads to a consistent upward trend in radiance across the WCO_2 band. This primarily can be attributed to enhanced scattering and absorption by aerosols, which elevate atmospheric path radiance. The effect of surface albedo is even more pronounced, producing a marked increase in radiance throughout the entire spectral window. As surface reflectance increases, more solar radiation is reflected upward from the surface, leading to a nearly linear enhancement in the observed radiance.

IV. Discussion

A. Surrogate Modeling of Jacobian Matrices

The Jacobian matrices refer to the partial derivatives of radiance with respect to each parameter in the state vector [17]. It quantifies the sensitivity of model outputs to perturbations in input parameters and plays a vital role in remote sensing retrievals [43], particularly in parameter optimization and uncertainty propagation. Accurate computation of Jacobian matrices can significantly improve the convergence efficiency and precision of retrieval algorithms, especially in atmospheric composition inversion tasks. However, conventional computation using high-fidelity RTM, such as SCIATRAN, is extremely time-consuming. The computational burden becomes particularly pronounced under conditions involving fine spectral resolution and multiparameter joint retrievals, often rendering such methods impractical for large-scale or near-real-time applications.

TABLE III
Accuracy Assessment of Jacobian Matrix Simulations

Parameter	Model	MAE	RMSE	R^2
XCO_2	AE	1.18E-05	2.83E-05	1.00
	Surrogate model	1.63E-04	4.57E-04	0.95
AOD	AE	4.28E-05	6.94E-05	1.00
	Surrogate model	1.97E-04	3.64E-04	0.99
Albedo	AE	4.19E-05	1.01E-04	1.00
	Surrogate model	4.39E-04	6.02E-04	1.00

Therefore, this study explores the surrogate model of Jacobian matrices for three representative parameters: XCO_2, AOD, and albedo. These variables represent the retrieval target (XCO_2), AOD, and lower boundary condition (albedo), encompassing the primary sources of sensitivity in retrieval problems. To evaluate the Jacobian matrices, the same network architecture as that of the radiance surrogate model is employed. The input features are identical to those used for radiance simulation, while the output layer is replaced with the corresponding Jacobian values with respect to XCO_2, AOD, and albedo. Each Jacobian component is trained separately using the SCIATRAN-simulated Jacobians, allowing the model to learn the nonlinear mapping between physical parameters and their differential radiative responses.

This study evaluates the simulation accuracy of Jacobian matrices for three parameters using both an AE and the surrogate model, as summarized in Table III. The AE performs only dimensionality reduction and reconstruction, and the surrogate model further incorporates the DNN-LSTM module. Results indicate that both models achieve satisfactory accuracy for all tested parameters, with the AE slightly outperforming the surrogate model in terms of statistical

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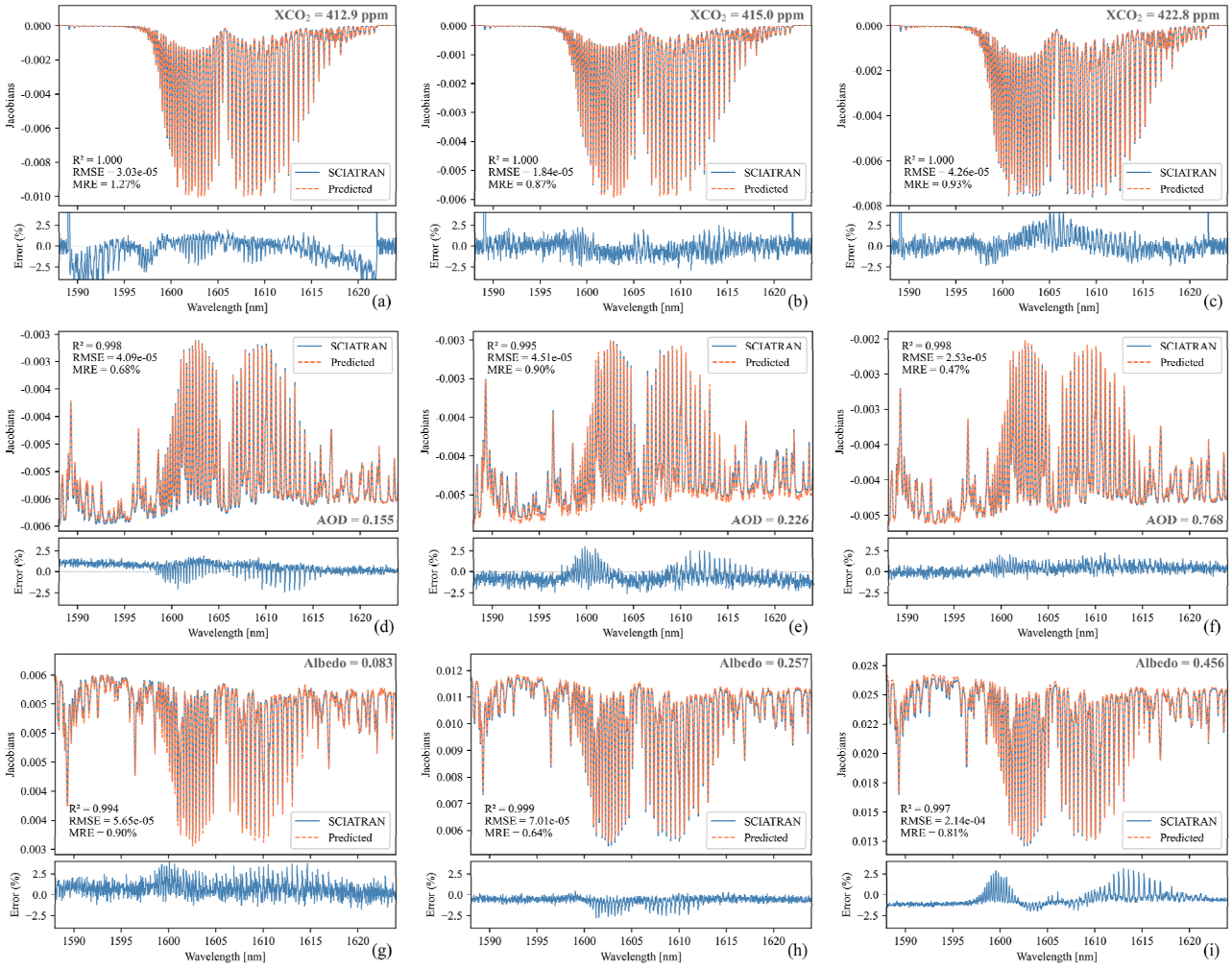

Fig. 9. Surrogate model predictions of Jacobian matrices. (a)–(c) Correspond to XCO_2, (d)–(f) to AOD, and (g)–(i) to albedo.

metrics. Meanwhile, we note that the accuracy of the XCO_2 Jacobian emulation is indeed lower than that for AOD or albedo. This may be because the WCO_2 band exhibits a more complex and nonlinear response to CO_2 variations. In the future, performance could be further improved by enlarging the training dataset with additional Jacobian samples generated specifically for XCO_2 perturbations. Fig. 9 presents a visual comparison between the Jacobian matrices generated by the surrogate model and those from the SCIATRAN reference across selected samples for all three parameters. The results demonstrate that the surrogate model reliably captures the spectral sensitivity to input perturbations with high fidelity. Overall, the surrogate model achieves a favorable balance between simulation accuracy and computational efficiency, significantly reducing the cost of Jacobian matrix estimation and providing a promising approach for efficient gradient-based optimization in atmospheric retrieval applications.

B. Preliminary Results of XCO_2 Retrieval Based on the Surrogate Model

To preliminarily evaluate the feasibility of the proposed radiative transfer surrogate model for XCO_2 retrieval, we embed both the spectral and Jacobian matrix surrogate models into the ACOS retrieval framework and design two experiments to perform the validation. In the first experiment, a single SCIATRAN-simulated radiance spectrum, randomly selected from the test set and excluded from the surrogate model's training data, is used as the retrieval target, with its corresponding input parameters treated as the reference truth. The priori values (XCO_2 = 413 ppm, AOD = 0.092, and albedo = 0.150) are randomly selected. In the other experiment, a single OCO-2 L1 observed spectrum is used as the target, while the corresponding OCO-2 L2 product values are regarded as reference truth. CAMS, SCIAMACHY surface Lambertian-equivalent reflectivity (LER) [44], and MOD04 5-Min L2 Swath 3-km data [45] are used as the prior values for XCO_2, albedo, and AOD, respectively, to retrieve XCO_2 for the case with sounding_ID 2020010118141433. It is worth noting that the primary objective of this study is to assess the feasibility of the AI-based surrogate model for radiative transfer forward simulation, and the validation of the retrieval process remains limited at this stage. The ACOS retrieval algorithm itself is highly complex, involving many uncertainties in aspects such as prior values and covariance matrices. Therefore, this study does not present large-scale

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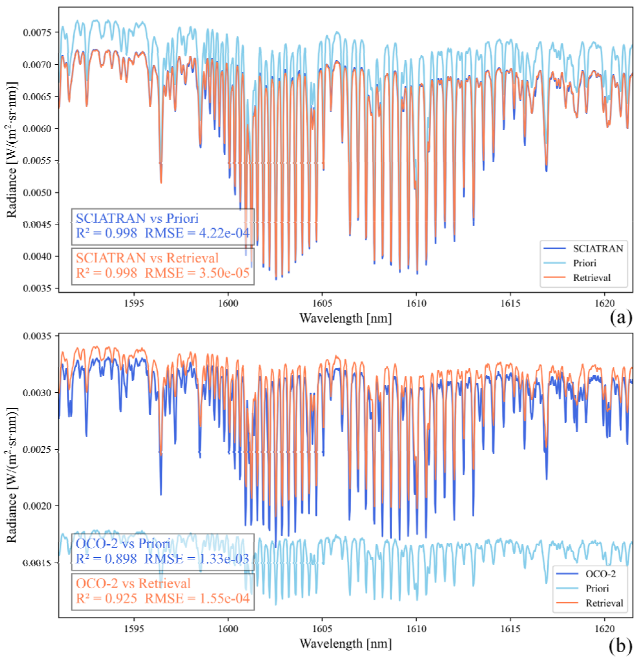

Fig. 10. Radiance spectrum comparison between surrogate model simulations using prior estimates and retrieved results and reference spectra from (a) SCIATRAN and (b) OCO-2.

TABLE IV
COMPARISON OF TRUE VALUES, PRIOR ESTIMATES, AND RETRIEVAL RESULTS FROM THE SURROGATE MODEL UNDER SCIATRAN AND OCO-2 CASES

	Type	XCO_2/ppm	AOD	Albedo
SCIATRAN case	True	414.000	0.102	0.140
	Priori	413.000	0.092	0.150
	Retrieval	413.679	0.097	0.141
OCO-2 case	True	408.299	0.077	0.048
	Priori	408.528	0.185	0.024
	Retrieval	408.316	0.101	0.050

XCO_2 retrieval results. Through these preliminary experiments, we aim to explore the adaptability and potential of the surrogate model within the radiative transfer simulation and retrieval framework, laying a foundation for future research and practical applications.

The retrieval performance is evaluated using both synthetic spectra generated by SCIATRAN and real satellite observations from OCO-2, as summarized in Table IV. For the SCIATRAN-based case, the retrieved XCO_2 value (413.6792 ppm) closely matches the SCIATRAN input (414.000 ppm) and shows a clear improvement over the prior estimate (413.000 ppm). Similarly, the retrieved AOD and albedo also exhibit higher consistency with the true inputs. Fig. 10(a) further illustrates the spectral differences between the surrogate model outputs based on prior and retrieved parameters, benchmarked against the original SCIATRAN simulation. The RMSE of the retrieved spectrum reaches 3.50×10^{-5} W/(m^2·sr·nm), which is significantly lower than that of the prior spectrum 4.22×10^{-4} W/(m^2·sr·nm), demonstrating the model's ability to replicate forward simulations and support accurate retrievals of XCO_2, AOD, and albedo with high computational efficiency.

For the OCO-2 case, the retrieved XCO $_2$(408.316 ppm) is also closer to the observed value (408.299 ppm) than the prior estimate (408.528 ppm), confirming the model's capability in improving retrieval accuracy. However, due to the relatively low spectral sensitivity of XCO_2 in the near-infrared region, the retrieval performance remains sensitive to the accuracy of prior inputs. The retrieved AOD still exhibits a notable deviation from the OCO-2 record, suggesting that a single aerosol parameter (e.g., AOD) is insufficient to capture the complex scattering characteristics of atmospheric aerosols [46]. This underscores the current model's limitations in representing the detailed scattering behavior of aerosols and indicates the need for further improvement. In contrast, the albedo retrieval exhibits higher accuracy and a marked improvement over the prior estimate, highlighting both its stronger spectral sensitivity and the surrogate model's capability to capture surface reflectance features effectively. These findings are further supported by Fig. 10(b), where the RMSE is reduced from 1.33×10^{-3} to 1.55×10^{-4} W/(m^2·sr·nm) when using retrieved results, and R^2 increases from 0.898 to 0.925.

Building upon these case-wise experiments, we further validate the model's generalization performance through a batch retrieval over an OCO-2 satellite swath on January 2, 2020. As shown in Fig. 11, the spatial distribution of XCO_2 along the satellite swath indicates that the retrieval results are highly consistent with the OCO-2 observations, accurately capturing the latitudinal gradient and spatial variation of atmospheric CO_2. Fig. 11(b1) and (b2) presents the absolute differences between the prior and retrieved XCO_2 values relative to OCO-2 observations. Quantitatively, the retrieval results exhibit significant improvements, with R^2 increasing to 0.77, and the MAE and RMSE decreasing to 0.56 and 0.74 ppm, respectively. These values are markedly better than those of the prior estimates (R^2 = 0.69, MAE = 0.81 ppm, and RMSE = 1.04 ppm). These results further demonstrate the surrogate model's robustness across both synthetic and real-satellite conditions, offering a promising solution for efficient and accurate atmospheric retrievals in large-scale or near-real-time applications.

C. Computational Efficiency

To evaluate the computational efficiency of the surrogate model, a series of benchmark tests is conducted on a desktop computer equipped with an Intel Core i7-14700KF processor (3.40 GHz) and 16 GB of RAM. The results show that the traditional SCIATRAN model requires approximately 3 min to simulate a single radiance spectrum for the WCO_2 band (1588–1624 nm, with a spectral resolution of 0.025 nm). By comparison, the proposed surrogate model demonstrates a significant computational advantage—achieving batch simulation of 2754 spectra in just 1.1 s, corresponding to millisecond-level inference time per spectrum. This remarkable speedup is primarily attributed to the efficient forward inference of the NN

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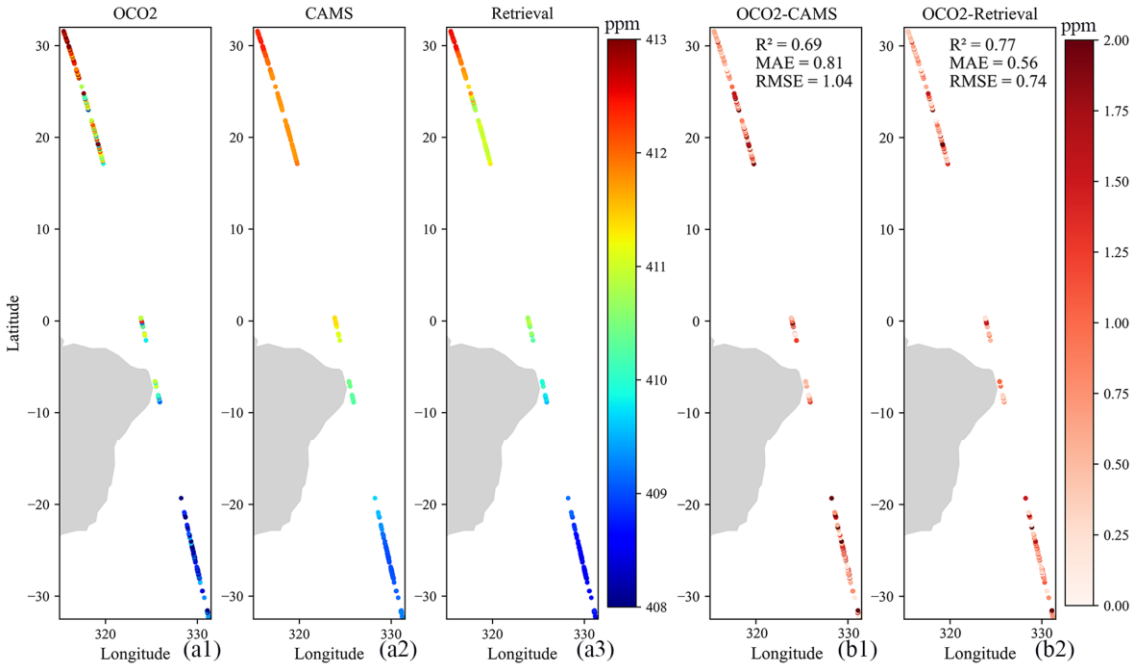

Fig. 11. Spatial distribution and retrieval evaluation of XCO_2 over an OCO-2 satellite swath on January 2, 2020. (a1)–(a3) XCO_2 from OCO-2 observations, CAMS prior estimates, and the retrieval results using the surrogate-model-integrated ACOS framework, respectively. (b1)–(b2) Errors between CAMS and OCO-2, and between the retrieval and OCO-2.

architecture employed in the surrogate model, which enables fast generation of large-scale spectral data while maintaining high accuracy.

In addition to forward simulation, the surrogate model also achieves significant acceleration in the retrieval process. When replacing the RTM in the ACOS framework with SCIATRAN, the two case studies presented in Section IV-B, namely, one based on SCIATRAN-simulated spectrum and the other on real OCO-2 observations, required approximately 29 and 26 min, respectively. By comparison, when using the ACOS embedded with both spectral and Jacobian surrogate models, the entire retrieval process can be completed within just 1.2 s per case. All these results indicate that the AI surrogate model has exceptionally high computational efficiency, making it an effective tool for large-scale and near-real-time inversion of XCO_2. In particular, it shows great promise in meeting the inversion demands of next-generation wide-swath carbon satellites with massive observational data.

D. Limitations and Future Work

Although this study has achieved preliminary progress in developing a surrogate model for radiative transfer and applying it to XCO_2 retrieval, several limitations remain and warrant further exploration in future work. This study primarily focuses on radiative transfer forward simulation in the WCO_2 band (1.6 μm), whereas the spectral features in the SCO_2 band (2.0 μm) and the O_2A band (0.76μ m) are also critically important for XCO_2 retrieval in practical applications [12], [47], [48]. Future research should, therefore, extend the framework to support multiband joint simulation, which is expected to improve retrieval robustness and reduce parameter uncertainty. In addition, due to the high complexity and variability of RTM input configurations, existing AI-based surrogate models still face challenges in generalizing across diverse atmospheric conditions, surface properties, and aerosol distributions. Enhancing the model's adaptability and generalization capability remains a key issue. Last, this study emphasizes the application of an AI-based surrogate model for SCIATRAN, significantly enhancing computational efficiency without improving retrieval accuracy. Future research will focus on hybrid modeling strategies that embed physical constraints into machine learning frameworks, aiming to improve both retrieval accuracy and computational efficiency.

V. Conclusion

Satellite remote sensing has become a crucial approach for monitoring atmospheric XCO_2. However, fully physical retrieval methods, particularly those based on RTM, suffer from high computational complexity and limited efficiency, posing significant challenges for large-scale and near-real-time retrieval applications. In contrast, purely data-driven AI retrievals have demonstrated remarkable efficiency and flexibility but often lack physical interpretability and uncertainty quantification, limiting their reliability in operational systems. To bridge this gap, this study proposes a deep learning-based surrogate model for the SCIATRAN in the WCO_2 band, aiming to preserve the high physical accuracy of physics-based models while significantly improving forward simulation efficiency. A large-scale synthetic dataset covering diverse combinations of atmospheric and surface parameters is generated to address the variability of real-world observational conditions. The developed surrogate model demonstrates excellent performance across multiple evaluation settings,

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achieving R^2 values close to 1.00 and RMSE values of 4.21×10^{-4}, 3.89×10^{-4}, and 2.87×10^{-4} W/(m^2·sr·nm) on the test datasets, finer-step extrapolation dataset, and real OCO-2 observation samples, respectively. In terms of computational speed, the model achieves millisecond-level inference for radiative transfer forward simulation, significantly outperforming conventional RTMs. Furthermore, this study verifies the surrogate model's applicability for Jacobian matrix simulation. The model accurately captures the sensitivities to key state parameters (XCO_2, AOD, and albedo), with R^2 values of 0.95, 0.99, and 1.00, respectively. Given that Jacobian matrices play a crucial role in retrieval by quantifying the spectral response to parameter perturbations, both the spectral and Jacobian matrix surrogate models are embedded into the ACOS retrieval framework. Preliminary experiments confirm their feasibility and effectiveness within retrieval workflows. In summary, the proposed surrogate model not only significantly accelerates radiative transfer forward simulation but also enables efficient estimation of Jacobian matrices for XCO_2 retrieval. This provides a solid technical foundation for high-efficiency and high-accuracy XCO_2 retrieval, positioning the model as a promising tool to support large-scale and near-real-time atmospheric retrieval in future satellite remote sensing missions.

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Ruohua Hu received the B.S. degree in remote sensing science and technology from Sun Yat-sen University, Zhuhai, China, in 2024, where she is currently pursuing the M.S. degree in geospatial engineering and science.

Her research interests include artificial intelligence and atmospheric CO_2 retrieval.

Tongwen Li (Member, IEEE) received the B.S. degree in geographic information systems and the Ph.D. degree in cartography and geographic information engineering from Wuhan University, Wuhan, China, in 2015 and 2020, respectively.

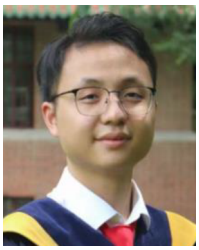
He is currently an Associate Professor with the School of Geospatial Engineering and Science, Sun Yat-sen University, Zhuhai, China. He has published more than 50 papers in peer-reviewed international journals. His research interests include atmospheric remote sensing (including fine particulate matter, ozone, carbon dioxide, and so on) and machine learning in environmental remote sensing.

Jingan Wu received the B.S. degree in geographic information systems from Anhui University, Hefei, China, in 2015, and the Ph.D. degree in cartography and geographic information engineering from Wuhan University, Wuhan, China, in 2021.

He is currently an Associate Researcher with the School of Geospatial Engineering and Science, Sun Yat-sen University, Zhuhai, China. His research interests include spatiotemporal data fusion and missing information reconstruction for remote sensing images.

Yuan Wang received the B.S. degree in 2017 and the Ph.D. degree from Wuhan University, Wuhan, China, in 2023.

He is currently a Professor with the School of Atmospheric Physics, Nanjing University of Information Science and Technology, Nanjing, China. His research interests include atmospheric environment remote sensing, artificial intelligence, and multifactor compound event assessment in atmospheric environments.



Lingfeng Zhang received the B.S. degree in remote sensing science and technology from Chengdu University of Information Technology, Chengdu, China, in 2021, and the M.S. degree in resources and environment engineering from the School of Geospatial Engineering and Science, Sun Yat-sen University, Zhuhai, China, in 2024. He is currently pursuing the Ph.D. degree in atmospheric science with Fudan University, Shanghai, China.

His research interests include deep learning, data reconstruction, retrieval algorithms of greenhouse gases, and CO_2 emission.

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